

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Bayesian Networks and Conditional Independence

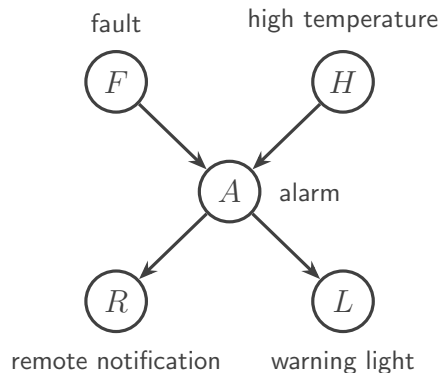
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September 15, 2026

More Evidence, More Variables



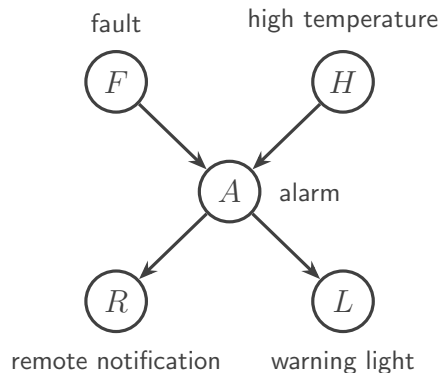
The Sensor Example

An alarm can reflect a fault or high temperature. The alarm may produce a notification and a warning light.

Representation Challenge

How can we specify a joint distribution over all five variables without listing every outcome?

More Evidence, More Variables



The Sensor Example

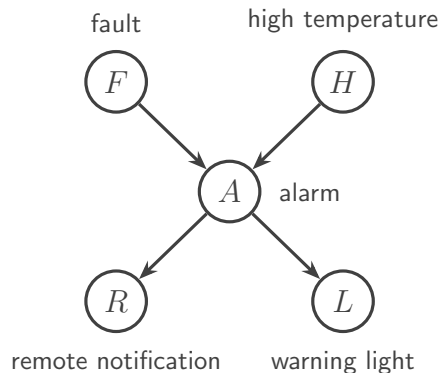
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Representation Challenge

How can we specify a joint distribution over all five variables without listing every outcome?

A Bayesian network combines a graph with local probability distributions.

Graph, Variables, and Parents



- ▶ Each node represents a random variable.
- ▶ The graph is directed and acyclic: no directed path returns to its starting node.
- ▶ A node's parents point directly into it:
 $\text{Parents}(A) = \{F, H\}$,
 $\text{Parents}(R) = \{A\}$.

All five variables are Boolean. Lowercase denotes true; for example, f is $F = \text{true}$ and $\neg f$ is $F = \text{false}$.

Local Probability Tables

Roots and Alarm

$$P(f) = 0.01, \quad P(h) = 0.10.$$

F	H	$P(a \mid F, H)$
true	true	0.99
true	false	0.89
false	true	0.41
false	false	0.01

Notification and Light

A	$P(r \mid A)$	$P(l \mid A)$
true	0.80	0.90
false	0.10	0.05

A conditional probability table (CPT) gives a distribution for each parent assignment.

The omitted Boolean probability is the complement:

$$P(\neg a \mid f, h) = 1 - 0.99 = 0.01.$$

The Same Alarm Model

The network assumes $F \perp\!\!\!\perp H$. Summing over the unobserved temperature gives the alarm rates from the first deck:

$$\begin{aligned} P(a \mid f) &= \sum_{h' \in \{h, \neg h\}} P(a \mid f, h') P(h') \\ &= 0.99(0.10) + 0.89(0.90) = 0.90, \end{aligned}$$

$$P(a \mid \neg f) = 0.41(0.10) + 0.01(0.90) = 0.05.$$

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$$P(a \mid \neg f) = 0.41(0.10) + 0.01(0.90) = 0.05.$$

The Previous Posterior Still Holds

$$P(f \mid a) = \frac{0.90(0.01)}{0.90(0.01) + 0.05(0.99)} = \frac{2}{13} \approx 15.4\%.$$

From the Chain Rule to Local Factors

For any distribution, the chain rule gives

$$P(F, H, A, R, L) = P(F)P(H \mid F)P(A \mid F, H)P(R \mid F, H, A)P(L \mid F, H, A, R).$$

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The network's independence assumptions reduce these conditionals to parent sets:

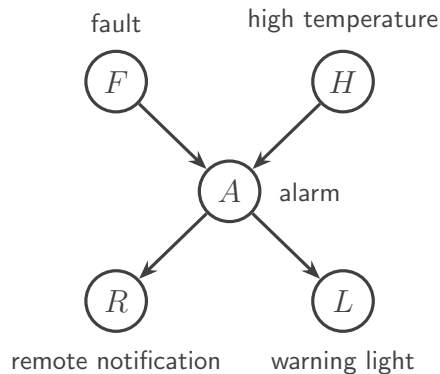
$$P(F, H, A, R, L) = P(F)P(H)P(A \mid F, H)P(R \mid A)P(L \mid A).$$

Bayesian-Network Factorization

For variables $X_1, \dots, X_n$ in a directed acyclic graph,

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i)).$$

One Complete Assignment

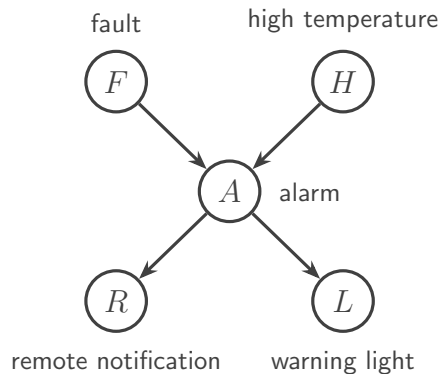


Event

A fault occurs, temperature is not high, and the alarm, notification, and warning light are all on.

$$P(f, \neg h, a, r, l) = P(f)P(\neg h)P(a \mid f, \neg h)P(r \mid a)P(l \mid a)$$

One Complete Assignment



Event

A fault occurs, temperature is not high, and the alarm, notification, and warning light are all on.

$$\begin{aligned} P(f, \neg h, a, r, l) &= P(f)P(\neg h)P(a \mid f, \neg h)P(r \mid a)P(l \mid a) \\ &= 0.01(0.90)(0.89)(0.80)(0.90) = 0.0057672 \approx 0.577\%. \end{aligned}$$

Ten Parameters Instead of Thirty-One

Unrestricted Joint Distribution

Five Boolean variables have $2^5 = 32$ assignments. Normalization removes one free parameter:

$$2^5 - 1 = 31.$$

This Bayesian Network

Node	Parents	Free parameters
F	0	1
H	0	1
A	2	4
R	1	2
L	1	2
Total		10

A Boolean node with k Boolean parents needs 2^k free CPT parameters. The savings come from the independence assumptions.

Parent Order and Local Independence

Construction in a Topological Order

1. Choose an order in which parents precede children, such as F, H, A, R, L .
2. For each variable, select earlier variables sufficient to describe its conditional distribution.
3. Specify its distribution given those parents.

Local Markov Property

A variable is conditionally independent of its non-descendants other than its parents, given its parents.

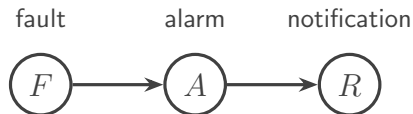
$$R \perp\!\!\!\perp \{F, H, L\} \mid A.$$

Different variable orders can represent the same distribution with different graph sizes.

Chain: Evidence on the Middle Node

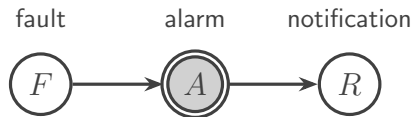
Before Observing the Alarm

Information can pass along the chain. A fault changes the chance of a notification.



Shading and a double border mark observed variables.

Chain: Evidence on the Middle Node



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Before Observing the Alarm

Information can pass along the chain. A fault changes the chance of a notification.

After Observing the Alarm

The middle node blocks this path:

$$F \perp\!\!\!\perp R \mid A.$$

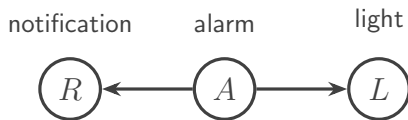
For either fault state,

$$P(r \mid a, F) = P(r \mid a) = 0.80.$$

Fork: Evidence on the Shared Parent

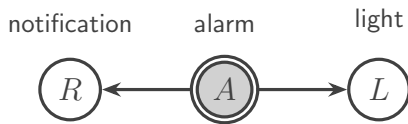
An Unobserved Shared Parent

A notification can provide evidence about the alarm and therefore about the warning light.



Shading and a double border mark observed variables.

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An Observed Shared Parent

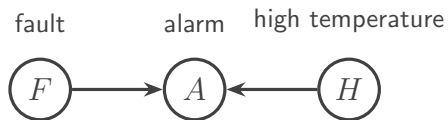
The middle node blocks this path:

$$R \perp\!\!\!\perp L \mid A.$$

Given that the alarm is on,

$$P(r, l \mid a) = 0.80(0.90) = 0.72.$$

Collider: A Shared Child



Two Arrowheads Meet

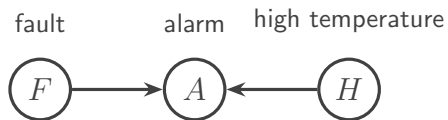
A is a collider on the path $F \rightarrow A \leftarrow H$.

No Evidence at the Collider or Its
Descendants

This path is blocked. The model gives

$$F \perp\!\!\!\perp H, \quad P(f \mid h) = P(f) = 0.01.$$

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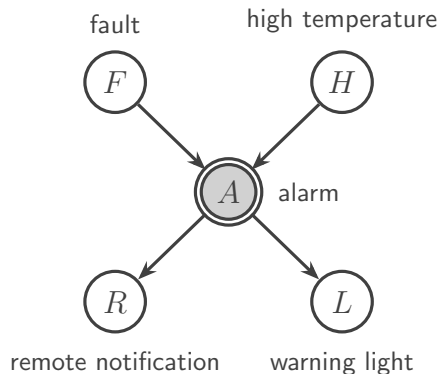
No Evidence at the Collider or Its
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This path is blocked. The model gives

$$F \perp\!\!\!\perp H, \quad P(f \mid h) = P(f) = 0.01.$$

Observing the collider, or one of its descendants, can activate this path.

Explaining Away: A Prediction



First Observation

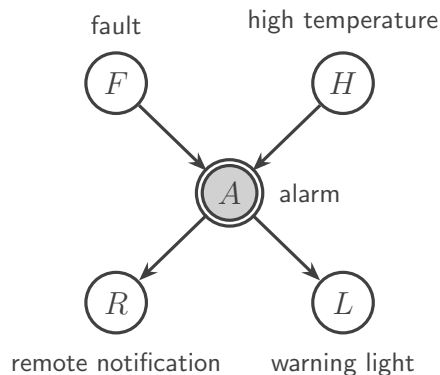
The alarm is on:

$$P(f \mid a) \approx 15.4\%.$$

A Second Observation

Temperature is high. Should the probability of a fault rise, fall, or stay the same?

Explaining Away: A Prediction



First Observation

The alarm is on:

$$P(f \mid a) \approx 15.4\%.$$

A Second Observation

Temperature is high. Should the probability of a fault rise, fall, or stay the same?

High temperature offers another explanation for the observed alarm. In this model, the fault probability falls.

Explaining Away: The Calculation

Alarm Alone

$$P(f \mid a) = \frac{0.009}{0.0585} \approx 15.4\%.$$

Alarm and High Temperature

Since $F \perp\!\!\!\perp H$, $P(f \mid h) = P(f) = 0.01$.

$$\begin{aligned} P(f \mid a, h) &= \frac{P(a \mid f, h)P(f)}{P(a \mid h)} \\ &= \frac{0.99(0.01)}{0.99(0.01) + 0.41(0.99)}. \end{aligned}$$

Explaining Away: The Calculation

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$$P(f | a) = \frac{0.009}{0.0585} \approx 15.4\%.$$

Alarm and High Temperature

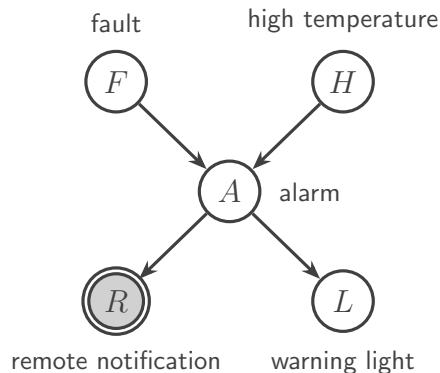
Since $F \perp\!\!\!\perp H$, $P(f | h) = P(f) = 0.01$.

$$\begin{aligned} P(f | a, h) &= \frac{P(a | f, h)P(f)}{P(a | h)} \\ &= \frac{0.99(0.01)}{0.99(0.01) + 0.41(0.99)}. \end{aligned}$$

$$P(f | a, h) = \frac{1}{42} \approx 2.38\%.$$

The fault and temperature are independent before observing the alarm, but dependent after observing it. Explaining away reduces the fault probability without making it zero.

Evidence at a Collider's Descendant

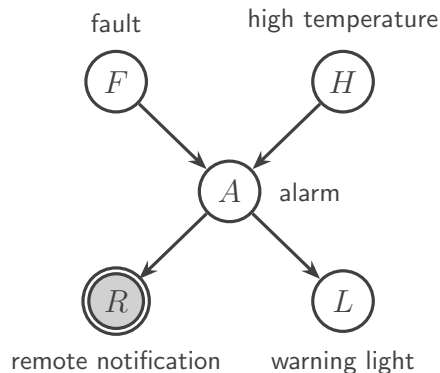


Only the Notification Is Observed

R is a descendant of the collider A on $F \rightarrow A \leftarrow H$.

- Observing R gives indirect evidence about A .
- The collider path becomes active even though A is unobserved.

Evidence at a Collider's Descendant



Only the Notification Is Observed

R is a descendant of the collider A on $F \rightarrow A \leftarrow H$.

- Observing R gives indirect evidence about A .
- The collider path becomes active even though A is unobserved.

The graph does not imply $F \perp\!\!\!\perp H \mid R$.
Dependence is possible; these CPTs produce it.

D-Separation: All Paths Matter

To test $X \perp\!\!\!\perp Y \mid Z$, treat paths as undirected when tracing connectivity. Assume X , Y , and the observed set Z are disjoint.

A Non-Collider on a Path

$$\rightarrow M \rightarrow \quad \leftarrow M \rightarrow \quad \leftarrow M \leftarrow$$

The path is blocked if $M \in Z$.

A Collider on a Path

$$\rightarrow M \leftarrow$$

The path is blocked unless M or at least one descendant of M is in Z .

D-Separation: All Paths Matter

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A Non-Collider on a Path

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Decision

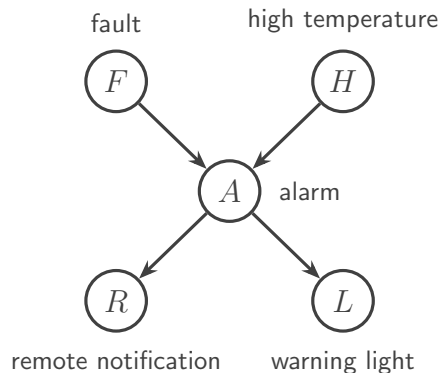
If every path from X to Y is blocked, they are d-separated, so $X \perp\!\!\!\perp Y \mid Z$. An active path permits dependence; it does not guarantee it for every choice of CPTs.

A Collider on a Path

$$\rightarrow M \leftarrow$$

The path is blocked unless M or at least one descendant of M is in Z .

Graph-Reading Checkpoint



Which statements follow from the graph? For each, identify the path and its blocking or activating evidence.

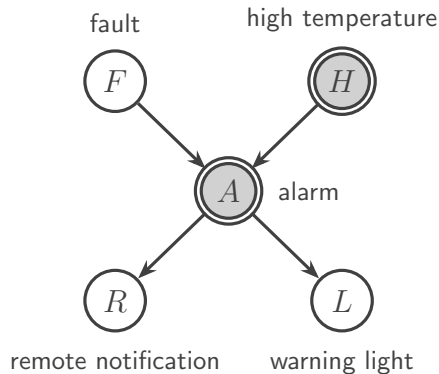
1. $F \perp\!\!\!\perp H$
2. $F \perp\!\!\!\perp H \mid A$
3. $R \perp\!\!\!\perp L \mid A$
4. $F \perp\!\!\!\perp L \mid R$
5. $F \perp\!\!\!\perp L \mid A, R$

Graph-Reading Answers

Statement	Implied?	Reason
$F \perp\!\!\!\perp H$	Yes	The unobserved collider $F \rightarrow A \leftarrow H$ blocks the path.
$F \perp\!\!\!\perp H \mid A$	No	Observing collider A activates the path.
$R \perp\!\!\!\perp L \mid A$	Yes	Observing the fork's middle node blocks $R \leftarrow A \rightarrow L$.
$F \perp\!\!\!\perp L \mid R$	No	The chain $F \rightarrow A \rightarrow L$ remains active; R is off this path.
$F \perp\!\!\!\perp L \mid A, R$	Yes	Observing non-collider A blocks $F \rightarrow A \rightarrow L$.

Each pair here has one undirected path. In larger graphs, one blocked path is insufficient if another remains active.

The Markov Blanket



A node's Markov blanket consists of its parents, its children, and its children's other parents.

For the Fault

$$\text{MB}(F) = \{A, H\}, \quad F \perp\!\!\!\perp \{R, L\} \mid A, H.$$

Observing child A activates the path to H , so H must also be in the blanket.

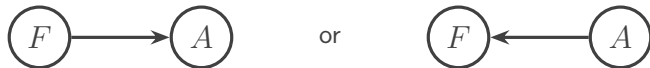
For the Alarm

$$\text{MB}(A) = \{F, H, R, L\}.$$

Arrow Direction and Causal Meaning

Probabilistic Meaning

An arrow identifies a parent used in a node's conditional distribution. The graph states a factorization and conditional-independence assumptions.



$$P(F, A) = P(F)P(A | F) = P(A)P(F | A).$$

Causal Meaning Requires More

With suitable CPTs, either two-node graph can represent the same joint distribution. Causal claims require additional assumptions about the system and interventions.

Representation, Independence, and Inference

- ▶ Local factors and a directed acyclic graph specify the full joint distribution.
- ▶ Missing edges express assumptions that can make the representation compact.
- ▶ Conditioning blocks chains and forks but can activate colliders.
- ▶ D-separation tests all paths to identify graph-implied independences.

Next: Exact Inference

We can multiply factors for a complete assignment. How can we efficiently answer a query such as $P(F \mid r, l)$ when H and A are unobserved?

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.